

Differential Privacy and Machine Unlearning

A Hypothesis Testing Approach

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Outline of the thesis

- **Chapter 1** : Introduction ([The Hypothesis Testing Approach](#))
- **Chapter 2** : Infinitely Divisible Privacy and Beyond ([Differential Privacy](#))
- **Chapter 3** : Gaussian Certified Unlearning ([Machine Unlearning](#))
- **Chapter 4** : Statistical Unlearning of Distributions ([Distributional Unlearning](#))
- **Chapter 5** : Conclusion and Future Directions

Research conducted during Ph.D.: thesis chapters and other projects

Projects included in the thesis

Chapter 2

Infinitely Divisible Privacy and Beyond I: Resolution of the $s^2 = 2k$ Conjecture with A. Maleki and S. Kulkarni, In submission

Chapter 3

Gaussian Certified Unlearning in High Dimensions: A Hypothesis Testing Approach, with A. Auddy, H. Zou, A. Maleki, and S. Kulkarni, ICLR 2026 Oral

Chapter 4

Statistical Unlearning of Distributions: A Hypothesis Testing Approach, with S. Kulkarni, In review

Projects not included

Community Detection in the Hypergraph Stochastic Block Model and Reconstruction on Hypertrees, with Y. Gu, COLT 2024

Exact Recovery in the Gaussian Weighted Stochastic Block Model: Information-Theoretic Thresholds, with S. Kulkarni, In review

Outline of the presentation

- Introduction ([The Hypothesis Testing Approach](#), 3 slides)
- Infinitely Divisible Privacy and Beyond ([Differential Privacy](#), 13 slides)
- Gaussian Certified Unlearning ([Machine Unlearning](#), 9 slides)
- Statistical Unlearning of Distributions ([Distributional Unlearning](#), 9 slides)

A closer look at the three problems

Differential privacy

Neighboring datasets

D vs. D'

Outputs $M(D), M(D')$

Goal: hard to distinguish
between the two inputs

Machine unlearning

A trained model $\hat{\beta}$

Users request removal

Retrain vs. unlearn

Goal: hard to distinguish
between the two inputs

Statistical unlearning

Harry Potter vs. Shakespeare

Carefully choose a subset
of Harry Potter database

Goal: unlearn Harry Potter
and retain Shakespeare

How well can one distinguish the inputs given the output?

The thesis in a couple of sentences

Differential privacy, machine unlearning, and statistical unlearning
are constraints that can be formalized through

indistinguishability.

We develop statistical foundations for these questions through

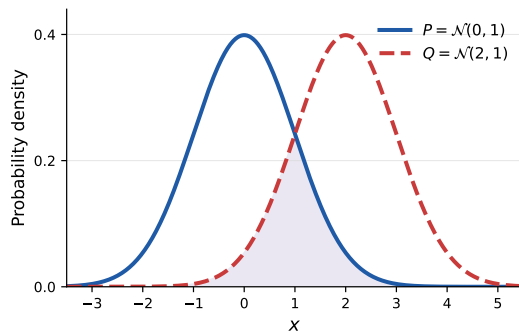
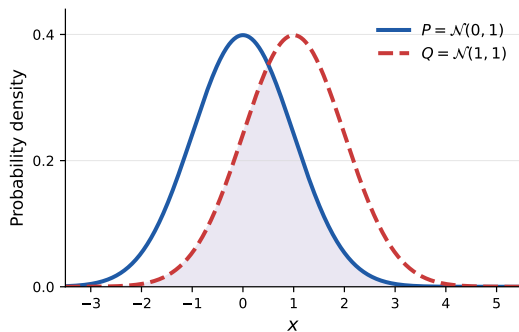
hypothesis testing.

Chapter 1

The Hypothesis Testing Approach

A pair of hypothesis testing problems

- Suppose one has a pair of hypothesis testing problems $\{P_0, Q_0\}$ and $\{P_1, Q_1\}$.
- **Question:** Which one among $\{P_0, Q_0\}$ and $\{P_1, Q_1\}$ is **harder to distinguish**?
- **Example:** $P_0 = \mathcal{N}(0, 1)$, $Q_0 = \mathcal{N}(1, 1)$, and $P_1 = \mathcal{N}(0, 1)$, $Q_1 = \mathcal{N}(2, 1)$ on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$.



Indistinguishability through Blackwell ordering

- **Step I :** We introduce the Neyman-Pearson curve (trade-off function).

$$f(\alpha) = T(P, Q)(\alpha) = \inf\{\mathcal{Q}(1 - \varphi) : \mathcal{P}(\varphi) \leq \alpha, \varphi : (\Omega_2, \mathcal{F}_2) \xrightarrow{m} [0, 1]\} \text{ for } \alpha \in [0, 1].$$

- **Step II:** Look at all the Neyman-Pearson curves by varying P, Q and $(\Omega, \mathcal{F})$.

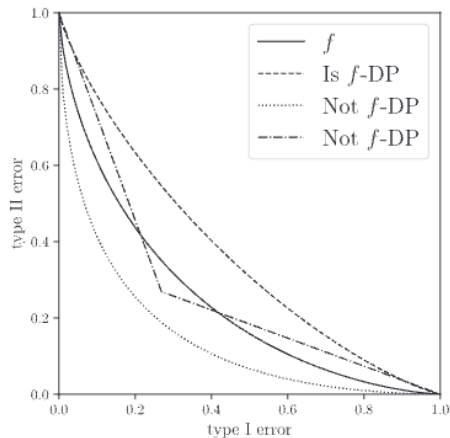
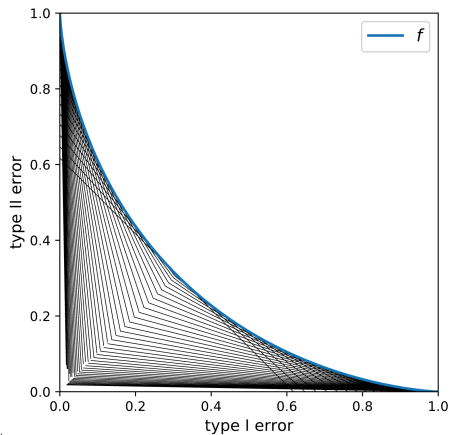
$$\mathcal{T} := \{T(P, Q) : [0, 1] \rightarrow [0, 1] : \{P, Q\} \text{ on } (\Omega, \mathcal{F})\}.$$

$$\mathcal{T} = \{f : [0, 1] \rightarrow [0, 1] : \text{convex, continuous, non-increasing, and } f(x) \leq 1 - x\}.$$

- **Step III:** Check whether $T(P_0, Q_0)(\alpha) \geq T(P_1, Q_1)(\alpha)$ on $[0, 1]$.

A higher $T(P, Q)$ curve means P and Q are harder to distinguish.

Blackwell ordering in pictures



Chapter 2

Differential Privacy

Joint work with Arian Maleki and Sanjeev Kulkarni

Differential privacy in reality

What can go wrong without differentially private data analysis?

Even if you release only “aggregates”, an attacker can attempt *membership inference*:

H_0 : Alice not included vs H_1 : Alice included.

The promise of differentially private machine learning

Changing **one persons record** changes the distribution of the released **output only slightly**.

Changing **one edge of a graph** changes the distribution of the released **output only slightly**.

For every pair of neighboring datasets D vs. D' , given the output $M(\cdot)$ of an algorithm, it should be hard to distinguish between the inputs D and D' .

Differential privacy through hypothesis testing

- Let $\Omega_1 \xrightarrow{M} \Omega_2$ be a **mechanism** (or more precisely a Markov kernel).
- **Abstractly**, we need to define a symmetric neighboring relation $w \sim w'$ on Ω_1 .
- **Example I**: $\Omega_1 = \chi^k$ with $S \sim S' \iff \Delta(S, S') := \sum_{i=1}^k \mathbf{1}(S_i \neq S'_i) \leq 1$.
- **Example II**: $\Omega_1 = \text{graphs}$ with $G \sim G' \iff \text{remove or add an edge}$.

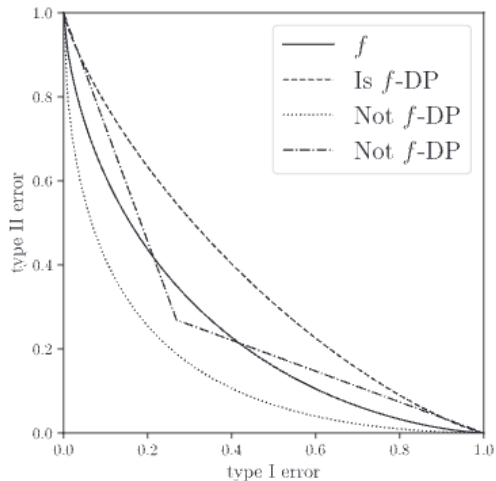
Definition (Dong-Roth-Su, f -differential privacy)

A Mechanism $M : (\Omega_1, \sim) \rightarrow \Omega_2$ is said to satisfy f -DP (for some TOF f) if

$$\inf_{S \sim S'} T(M(S), M(S')) \geq f \text{ pointwise everywhere on } [0, 1].$$

- $f = G_\theta = T(N(0, 1), N(\theta, 1))$ corresponds to Gaussian differential privacy.

Differential privacy through pictures



What happens when we release multiple statistics?

- For $f = T(P_1, Q_1), g = T(P_2, Q_2)$ define $f \otimes g = T(P_1 \otimes P_2, Q_1 \otimes Q_2)$.

Theorem (Composition theorem of Dong-Roth-Su)

If $(\Omega_1, \sim) \xrightarrow{M} \Omega_2$ is f -DP, and $(\Omega_1, \sim) \xrightarrow{N} \Omega_3$ is g -DP, then

$$\Omega_1 \xrightarrow{(M,N)} \Omega_2 \times \Omega_3 \text{ is } f \otimes g\text{-DP.}$$

- Repeated releases, therefore, lead to privacy degradation $f_n^{\otimes n} = T(P_n^{\otimes n}, Q_n^{\otimes n})$.

What non-trivial baselines appear in the limit of compositions?

A central limit theorem and Gaussian baselines

$$\text{kl}(f) := -\int_0^1 \log |f'(x)| dx, \quad \kappa_2(f) := \int_0^1 \log^2 |f'(x)| dx, \quad \kappa_3(f) := \int_0^1 |\log |f'(x)||^3 dx$$

Theorem (Central limit theorem of Dong-Roth-Su)

Let $\{f_{ni} : 1 \leq i \leq n\}_{n=1}^\infty$ be a triangular array of symmetric *trade-off functions* and assume the following limits for some constants $k \geq 0$ and $s > 0$ as $n \rightarrow \infty$:

- ① $\sum_{i=1}^n \text{kl}(f_{ni}) \rightarrow k$ and $\sum_{i=1}^n \kappa_2(f_{ni}) \rightarrow s^2$,
- ② $\max_{1 \leq i \leq n} \text{kl}(f_{ni}) \rightarrow 0$ and $\sum_{i=1}^n \kappa_3(f_{ni}) \rightarrow 0$.

$$\text{Then } \lim_{n \rightarrow \infty} f_{n1} \otimes f_{n2} \otimes \cdots \otimes f_{nn} = T(N(-k, s^2), N(k, s^2)).$$

- Miraculously, ‘*in all examples*’ we always have $s^2 = 2k$. ‘A proof of this interesting observation or the construction of a counterexample is left for future work’.

Poisson baselines

Dong-Roth-Su showed Gaussian limits for compositions of DP mechanisms.

$$\lim_{n \rightarrow \infty} f_{n1} \otimes \cdots \otimes f_{nn} = T \left(N \left(-\frac{s^2}{2}, s^2 \right), N \left(\frac{s^2}{2}, s^2 \right) \right) \text{ for some } s \in \mathbb{R}.$$

Lemma (Pandey-Maleki-Kulkarni)

If $f_{ni} = T \left(\text{Ber} \left(\frac{\lambda_1}{n} \right), \text{Ber} \left(\frac{\lambda_2}{n} \right) \right)$ for a pair of Bernoulli variables and $1 \leq i \leq n$, then

$$\lim_{n \rightarrow \infty} f_{n1} \otimes \cdots \otimes f_{nn} = T(P(\lambda_1), P(\lambda_2)).$$

What non-trivial baselines appear in the limit of compositions?

Infinitely divisible baselines in the limit of compositions

Theorem (Pandey-Maleki-Kulkarni)

Consider a sequence of trade-off functions

$$\{f_n = T(P_n, Q_n)\}_{n \geq 1}, \quad f_n(0) = 1 \quad \text{for every } n \geq 1.$$

If, for some trade-off function $f_\infty = T(P_\infty, Q_\infty)$,

$$f_n^{\otimes n}(\alpha) \longrightarrow f_\infty(\alpha) \quad \text{for every } \alpha \in [0, 1],$$

then f_∞ belongs to $\mathcal{I}_T$, where

$$\mathcal{I}_T = \left\{ f = T(P, Q) : P \text{ is infinitely divisible on } \mathbb{R}, dQ(x) = e^x dP(x) \right\}.$$

Gaussian and Poisson are Infinitely divisible

- **P infinitely divisible** $\leftrightarrow \log \hat{P}(t) = itm - \frac{1}{2}t^2\sigma^2 + \int_{\mathbb{R}} (e^{itx} - 1 - itx\mathbf{1}_{|x|\leq 1})d\nu(x)$, where ν a positive Borel measure on $\mathbb{R} \setminus 0$ satisfying $\int_{\mathbb{R}} (x^2 \wedge 1)d\nu(x) < \infty$.

$$P \stackrel{d}{=} \text{Gaussian part} + \text{Poisson part}$$

- **Poissons:** $f = T(P(\lambda_1), P(\lambda_2)) \in \mathcal{J}_T$ by first taking $N \stackrel{d}{=} P(\lambda_1)$ and then considering

$$P \stackrel{d}{=} \lambda_1 - \lambda_2 + N \log(\lambda_2/\lambda_1), \text{ and } dQ(x) = e^x dP(x).$$

Resolution of the $s^2 = 2k$ conjecture of Dong-Roth-Su

Dong-Roth-Su observed that Gaussian limits always have the following form

$$\lim_{n \rightarrow \infty} f_n^{\otimes n} = T(N(-k, s^2), N(k, s^2)) \text{ for some } s \in \mathbb{R} \text{ and } s^2 = 2k.$$

- **Dong-Roth-Su** asked whether $s^2 = 2k$ always holds in the limit of compositions?

The answer is **Yes**, always, as long as there is no Poisson part.

- Consider the limiting Gaussian case with $P \stackrel{d}{=} -k + |s|Z$.
- Since $Q(\mathbb{R}) = P(\exp x) = \exp(-k + s^2/2) = 1$ we have $s^2 = 2k$.

Gaussian mechanism

Theorem

Let $(\Omega_1, \sim) \xrightarrow{g} \mathbb{R}$ ‘*continuous*’, $w_g = \sup_{w \sim w'} |g(w) - g(w')|$ and $f_\infty = T(N(0, 1), N(\mu, 1))$. Define

$$M : \mathbb{R} \rightarrow \mathbb{R} \quad \text{as} \quad M(g(x)) \stackrel{d}{=} g(x) + cZ, \text{ for } Z \sim N(0, 1),$$

with $w_g = c\mu$, for $\mu > 0$. Then M satisfies μ -*Gaussian differential privacy*. Equivalently,

$$\inf_{w \sim w'} T(M(g(w)), M(g(w'))) = \min(f_\infty, f_\infty^{-1}).$$

- What if $g : (\Omega_1, \sim) \rightarrow \mathbb{Z}_{\geq 0}$ is ‘*discrete*’ and *privacy baseline* $f_\infty = T(P(\mu_1), P(\mu_2))$?

Construct a mechanism $M : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0}$ that satisfies *Poisson differential privacy*!

Poisson mechanism

Theorem (Pandey-Maleki-Kulkarni)

Let $(\Omega_1, \sim) \xrightarrow{g} \mathbb{Z}_{\geq 0}$ ‘discrete’ and $f_\infty = T(P(\mu_1), P(\mu_2))$ for $\mu_2 > \mu_1 > 0$. Define

$$M : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0} \text{ as } M(g(x)) \stackrel{d}{=} P(N_2 e^{N_1 g(x)}),$$

with $N_1 w_g = \log(\mu_2/\mu_1)$ and $N_2 w_{h \circ g} = |\mu_2 - \mu_1|$, where $h(y) = (\mu_2/\mu_1)^{y/w_g}$.

Then, M satisfies (μ_1, μ_2) Poisson differential privacy. Equivalently,

$$\inf_{w \sim w'} T(M(g(w)), M(g(w'))) \geq \min(f_\infty, f_\infty^{-1}).$$

$\geq$ becomes $=$ for $\Omega_1 = \{0, 1\}^{\binom{n}{2}}$ and g is monotone increasing and supermodular.

Summary of chapter 2

- We characterize the list of privacy baselines in the limit of composition.

$$\mathcal{I}_T = \{f = T(P, Q) : P \text{ is infinitely divisible on } \mathbb{R}, dQ(x) = e^x dP(x) \}.$$

- We describe a larger Gaussian + Poisson theory and resolve the $s^2 = 2k$ conjecture.

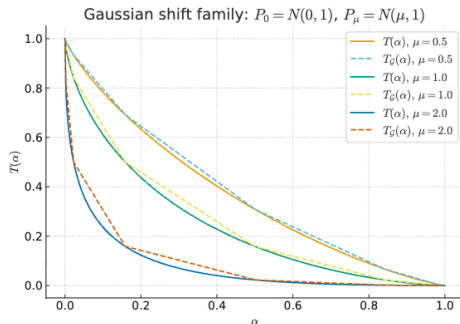
$$P \stackrel{d}{=} \textbf{Gaussian part} + \textbf{Poisson part}$$

- We provide a mechanism for count statistics under the Poisson baselines.

$$M : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0} \text{ as } M(g(x)) \stackrel{d}{=} P(N_2 e^{N_1 g(x)}).$$

Privacy beyond Infinite divisibility

- How do we incorporate **computation** into our framework of privacy?
- What happens when we shrink the σ -algebra $\mathcal{G} \subset \mathcal{F}$ for (P, Q) on $(\Omega, \mathcal{F})$?



- How do we incorporate **quantization** into our framework?

Chapter 3

Machine Unlearning

Joint work with Arnab Auddy, Haolin Zou, Arian Maleki, and Sanjeev Kulkarni

Machine unlearning: motivation and goal

• What is the motivation behind machine unlearning?

- Users may request that their personal records be deleted from trained models.
- Companies must delete both the data and its statistical influence from trained models.
- Re-training from scratch for every deletion request is prohibitively expensive.

• What is the goal of machine unlearning?

- Develop **efficient algorithms** to remove the influence of selected data points.
- It avoids full retraining while preserving **generalization capabilities** of the model.
- It allows **unlearning** by enabling the users to exercise their right to be forgotten.

Mathematical framework: dataset, training, and retraining

- **Dataset** $D_n = \{(x_i, y_i)\}_{i=1}^n$ consists of feature-response pairs $(x_i, y_i) \in \mathbb{R}^p \times \mathbb{R}$.
- We assume they are IID samples from a generalized linear model $p(x)q(y \mid x^\top \beta^*)$.
- We assume the learning algorithm is **Regularized empirical risk minimization**.

$$\text{RERM: } \hat{\beta} = A(\mathcal{D}_n) := \arg \min_{\beta \in \mathbb{R}^p} L(\beta) := \sum_{i=1}^n \ell(y_i, x_i^\top \beta) + \lambda r(\beta).$$

- The **ideal retraining** for a given deletion request $M \subset [n]$ with $|M| = m$ takes the form

$$\text{Retraining: } \hat{\beta}_{\setminus M} = \arg \min_{\beta \in \mathbb{R}^p} L_{\setminus M}(\beta) = \lambda r(\beta) + \sum_{i \notin M} \ell(y_i, x_i^\top \beta).$$

The unlearning algorithm must satisfy three goals

- **Unlearning** is a randomized procedure $\bar{A}$ having access to the trained model $\hat{\beta}$

$$\text{Unlearning: } \tilde{\beta}_{\setminus M} := \bar{A}(\hat{\beta}, D_M, T(D_n), b)$$

- We assume it has access to D_M , Hessian information $T(D_n)$, and external noise b .

$$\tilde{\beta}_{\setminus M} := \bar{A}(\hat{\beta}, D_M, T(D_n), b)$$

Efficient

Avoid optimization
from scratch.

Certified

Output should look like
exact retraining.

Accurate

Preserve predictive
performance.

Certifiability through hypothesis testing

- We **introduce f -certifiability** through a hypothesis testing problem for an adversary.
- An adversary observing the output $\tilde{\beta}_{\setminus M}$ tries to distinguish between the two.

$$H_0 : \tilde{\beta}_{\setminus M} \sim P_{\text{re}} \quad \text{vs.} \quad H_1 : \tilde{\beta}_{\setminus M} \sim P_{\text{un}}$$

- $P_{\text{re}} \stackrel{d}{=} \bar{A}(\hat{\beta}_{\setminus M}, \emptyset, T(D_{\setminus M}), b)$ is $\bar{A}$ run on $D_{\setminus M}$ with no deletion requests.
- $P_{\text{un}} \stackrel{d}{=} \bar{A}(\hat{\beta}, D_M, T(D), b)$ is $\bar{A}$ run on D with deletion request M .
- Certifiability holds if and only if the adversary cannot distinguish H_0 from H_1 .

Gaussian certifiability: a canonical choice in high-dimensions

Definition (f-certifiability, (Pandey et al.))

An algorithm $\bar{A}$ is (ϕ, f) -certifiable if with probability $\geq 1 - \phi$ over the data D

$$\inf_{|M| \leq m} \min(T(P_{\text{re}}, P_{\text{un}})(\alpha), T(P_{\text{un}}, P_{\text{re}})(\alpha)) \geq f(\alpha) \quad \forall \alpha \in [0, 1].$$

- Log-concave mechanisms “behave” like a scaled Gaussian mechanism as $p \uparrow \infty$.
- Gaussian mechanism is the canonical mechanism to achieve certifiability in HD.
- Gaussian certifiability with $f = f_{G,\varepsilon}$ *tightly* characterizes the Gaussian mechanism.

The Gaussian trade-off curve: $f_{G,\varepsilon}(\alpha) := T(N(0, 1), N(\varepsilon, 1))(\alpha)$

Accuracy through generalization error divergence

- An algorithm that outputs **pure noise achieves perfect certifiability but zero utility**.
- We need a criterion to measure the generalization capabilities of the unlearned model.
- We measure this for a fresh test point (x_0, y_0) independent of D (Zou et al. 2025):

$$\text{GED}_\ell(A, \bar{A}; M, D_n) := \mathbb{E} \left[\left| \ell(y_0, x_0^\top \hat{\boldsymbol{\beta}}_{\setminus M}) - \ell(y_0, x_0^\top \tilde{\boldsymbol{\beta}}_{\setminus M}) \right| \middle| D_n \right]$$

- $\text{GED} = 0$ means the unlearned model generalizes identically to the retrained model.

Newton-based unlearning algorithm

- Step 1 — Newton approximation.**

$$\hat{\boldsymbol{\beta}}_{\setminus M}^{(1)} = \hat{\boldsymbol{\beta}} - [\nabla^2 L_{\setminus M}(\hat{\boldsymbol{\beta}})]^{-1} \nabla L_{\setminus M}(\hat{\boldsymbol{\beta}})$$

- Since $\nabla L(\hat{\boldsymbol{\beta}}) = 0$, $\nabla L_{\setminus M}(\hat{\boldsymbol{\beta}}) = -\sum_{i \in M} \dot{\ell}(y_i, x_i^\top \hat{\boldsymbol{\beta}}) x_i$ involves only the deleted data.

- Step 2 — Gaussian noise injection.**

$$\tilde{\boldsymbol{\beta}}_{\setminus M} = \hat{\boldsymbol{\beta}}_{\setminus M}^{(1)} + b, \quad b \sim \mathcal{N}\left(0, \frac{R^2}{\varepsilon^2} I_p\right)$$

- We choose $R = C_1(n) \sqrt{C_2(n) m^3 / (2\lambda v n)}$ for some $C_1, C_2 = O(\text{polylog}(n))$.
- These constants in the noise calibration can be computed for a given problem.

Theoretical results: certifiability and accuracy (Pandey et al.)

Gaussian certifiability: Under some assumptions on ℓ , r , and D , the one-step noisy Newton $\tilde{\beta}_{\setminus M}$ with noise $b \sim \mathcal{N}(0, R^2 \varepsilon^{-2} I_p)$ achieves $(\phi_n, f_{G,\varepsilon})$ certifiability with

$$R = C_1(n) \sqrt{\frac{C_2(n)m^3}{2\lambda v n}}, \quad \phi_n = nq_n^{(y)} + 8n^{-3} + ne^{-p/2} + 2e^{-p} \rightarrow 0.$$

Vanishing GED: With the same assumptions and the choice of R as above, we have with probability $\geq 1 - (n+1)q_n^{(y)} - 14n^{-3} - ne^{-p/2} - 2e^{-p} - e^{-(1-\log 2)p}$:

$$\text{GED}(\tilde{\beta}_{\setminus M}, \hat{\beta}_{\setminus M}) \leq C_1(n) \sqrt{C_2(n)} \left(\frac{1}{\varepsilon} + \frac{1}{\sqrt{p}} \right) \sqrt{\frac{m^3(m+2)}{\lambda v n}} \cdot \text{polylog}(n).$$

Summary of chapter 3 and machine unlearning beyond Newton

- One noisy Newton step achieves both Gaussian certifiability and vanishing GED.
- Zou et al. showed that at least two Newton steps are required to achieve (ϕ, ϵ) -certifiability with vanishing GED. We show that one step achieves both.
- This is because (ϕ, ϵ) -certifiability is sub-optimal under noise adding mechanisms.
- Gaussian certifiability is the canonical notion of certifiability in high dimensions.
- The Gaussian mechanism is the canonical mechanism to achieve certifiability in HD.
- It would be interesting to extend the current framework with
 - Non-convex loss functions and first-order methods,
 - Online unlearning and going beyond the $p \sim n$ regime.

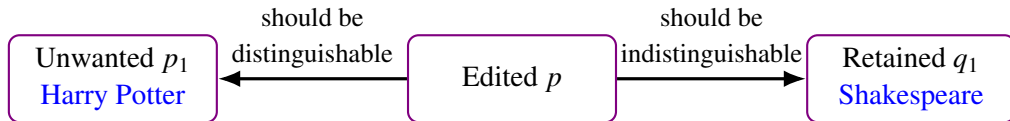
Chapter 4

Statistical Unlearning

Joint work with Sanjeev Kulkarni

From unlearning individual records to unlearning a distribution

- In machine unlearning, the list of individual data points to be removed is known.
- In statistical unlearning, one carefully selects a subset of the database to unlearn.



- We have **many** samples from the unwanted p_1 and the retained q_1 used in training.
- We have to **choose a subset** of the p_1 database to unlearn p_1 for any downstream task.
- **Full deletion** of p_1 can be computationally expensive for downstream unlearning.
- **Random partial removal** of p_1 may not provide distribution-level guarantees.

A two-sided hypothesis-testing definition

Definition (Pandey-Kulkarni)

Given the unwanted p_1 and the retained q_1 , baseline trade-off functions $f_d, f_c \in \mathcal{T}$, we say that an edited distribution $p \in \mathcal{P}$ satisfies (f_d, f_c) distributional unlearning if

$$T(p, p_1) \leq f_d \quad (\text{removal}) \quad T(p, q_1) \geq f_c \quad (\text{preservation}).$$

- Removal asks p to be **easy to distinguish** from the unwanted population p_1 .
- Preservation asks p to be **hard to distinguish** from the retained population q_1 .

Feasible region, downstream guarantee, and composition rules

Feasible region: $\mathcal{R} = \{p \in \mathcal{P} : T(p, p_1) \leq f_d, \text{ and } T(p, q_1) \geq f_c\}$

Lemma (Robust downstream generalization guarantee, Pandey-Kulkarni)

If $p \in \mathcal{R}$ with $f_d = T(P_d, Q_d)$, $f_c = T(P_c, Q_c)$ and $TV(P, Q) := \sup_{\|h\|_\infty \leq 1} |P(h) - Q(h)|$.
Then

$$TV(p, p_1) \geq TV(P_d, Q_d) \text{ and } TV(p, q_1) \leq TV(P_c, Q_c).$$

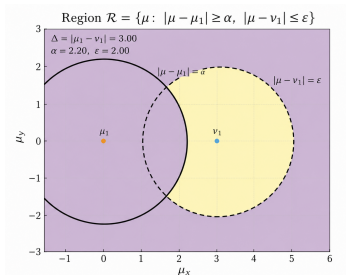
- We provide **composition rules** of statistical unlearning where one has to unlearn product distributions $p'_1 \otimes p''_1$ having multiple modes, such as text p'_1 and image p''_1 .

Feasible region of the Gaussian family

Lemma (Pandey-Kularni)

Let $p_1 = N(\mu_1, I)$, $q_1 = N(v_1, I)$, $\mathcal{P} = \{N(\mu, I) : \mu \in \mathbb{R}^d\}$ and $f_d = G_\alpha$, $f_c = G_\varepsilon$ for $\alpha, \varepsilon \geq 0$ and $G_\theta = T(N(0, 1), N(\theta, 1))$. Denote $\Delta = \|\mu_1 - v_1\|_2$. Then

$$\mathcal{R} = \{N(\mu, I) : \|\mu - \mu_1\|_2 \geq \alpha, \|\mu - v_1\|_2 \leq \varepsilon\}, .$$



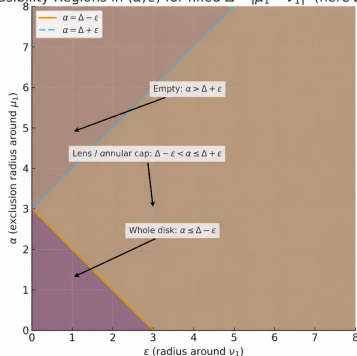
$\alpha \leq \Delta - \varepsilon : \mathcal{R} \equiv \text{whole ball,}$

$\Delta - \varepsilon < \alpha \leq \Delta + \varepsilon : \mathcal{R} \equiv \text{annular cap,}$

$\alpha > \Delta + \varepsilon : \mathcal{R} \equiv \emptyset.$

Removal-preservation Pareto frontier of the Gaussian family

Feasibility Regions in (α, ϵ) for fixed $\Delta = |\mu_1 - \nu_1|$ (here $\Delta = 3.0$)



Feasibility ($\mathcal{R} \neq \emptyset$) requires

$$0 \leq \alpha \leq \Delta + \epsilon.$$

For preservation level $\epsilon \geq 0$,

maximum removal level $\alpha = \Delta + \epsilon$.

Hence $\text{PF} = \{(\Delta + \epsilon, \epsilon) : \epsilon \geq 0\}$.

Feasible regions and Pareto frontiers beyond Gaussian

We derive analogous feasible-region and Pareto frontier descriptions for several $\mathcal{P}$.

Log-concave

One-parameter shifted log-concave family.

Gaussian white noise

Non-parametric regression with $\mu \leftrightarrow f \in L^2[0, 1]$.

Poisson family

A non-location family for unlearning through counts.

The hypothesis-testing formulation adapts the baselines (f_d, f_c) to the family $\mathcal{P}$.

Random removal versus scoring-based selective removal

We have n_1 samples from p_1 and n_2 samples from q_1 , and may delete n_r points from p_1 .

Random removal baseline

- 1 Choose n_r samples uniformly at random from p_1 to unlearn.
- 2 Fit the edited p by the weighted MLE on what remains.

Ignores the most harmful samples.

Scoring-based selective removal

- 1 Estimate the retained center $\hat{v}_1$ by MLE on n_2 samples from q_1 .
- 2 Score samples $\{x_i\}_i$ from p_1 by $s_i = \|x_i - \hat{v}_1\|_2$.

Deletes the n_r largest scores.

Finite samples reveal an information–computation gap

For isotropic shifted Gaussians, $\Delta = \|\mu_1 - v_1\|_2$. Information theoretically, allowable
(removal level, preservation level) = (α, ε) , with any $\varepsilon \geq 0$ and $0 \leq \alpha \leq \Delta + \varepsilon$.

Finite-sample procedures obey smaller Pareto frontiers and show an information-computation gap.

Theorem (Pareto frontier of algorithms, Pandey-Kulkarni)

Let $p_1 = N(\mu_1, I), q_1 = N(v_1, I) \in \mathcal{P} = \{N(\mu, I) : \mu \in \mathbb{R}^d\}$ and $\delta \in (0, 1)$. With probability $1 - \delta$ the edited data $p \in \mathcal{R}$ if

$$\text{Random removal: } \varepsilon \geq \varepsilon_m(R), \quad 0 \leq \alpha \leq \Delta - \varepsilon_m(R),$$

$$\text{Selective removal: } \varepsilon \geq \varepsilon_m(S), \quad 0 \leq \alpha \leq \Delta - \varepsilon_m(S).$$

Estimating population centers creates the gap. Under sufficient separation and deletion budget, we show $\varepsilon_m(S) \leq \varepsilon_m(R)$ and move closer to the information-theoretic Pareto frontier.



Summary of chapter 4 and future directions

- We introduce a framework of unlearning from selected records to **entire populations**.
- We introduce a two-sided testing criterion balancing **removal** and **preservation**.
- Characterize feasible regions and Pareto frontiers for **Gaussian** and **Poisson** models.
- We compare random removal and scoring based removal algorithms.
- It would be interesting to extend the current framework with
 - likelihood-based selection algorithms for broader parametric families,
 - unwanted labels are uncertain and mixed with the retained ones.

Overall message of the thesis

Whenever an algorithmic requirement can be stated as a claim about distinguishability, hypothesis testing provides a natural language in which to formulate and analyze it.

Thank you.

Questions?

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